- training error : $J(\theta) = \frac{1}{n} \sum_{i=1}^{n} (y^{(i)} - h_\theta(x^{(i)}))^2$
- test error : $L(\theta) = E_{(x,y)\sim P} (y - h_\theta(x))^2$

$\Rightarrow$ generalization gap

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underfitting : train ☹

overfitting : train ☺ test ☹

$f(x_0) + \varepsilon$ and $\hat{f}(x_0)$

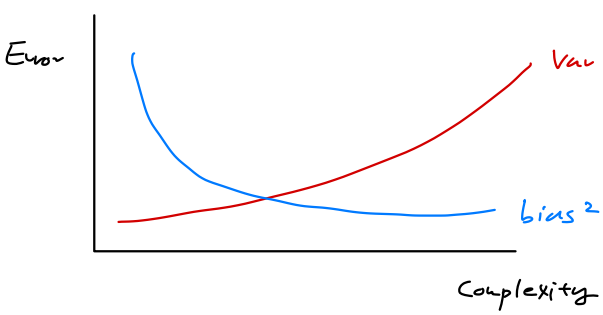
(i) $E[(f(x_0) - \hat{f}(x_0) + \varepsilon)^2] = E[(f(x_0) - \hat{f}(x_0))^2]$

$+ E[2\varepsilon (f(x_0) - \hat{f}(x_0))] + \underbrace{E[\varepsilon^2]}_{\text{irreducible}}$

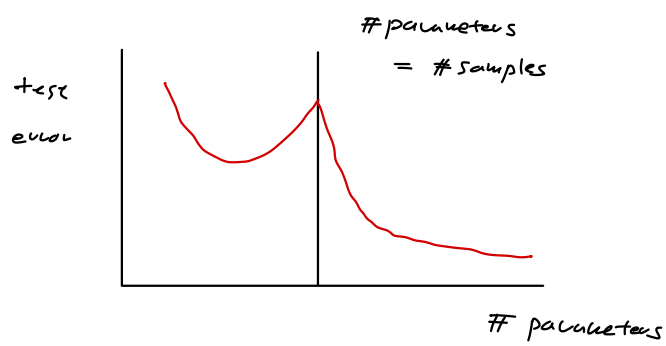
(ii) $E[(f(x_0) - \hat{f}(x_0))^2] = E[(f(x_0) - E[\hat{f}(x_0)])^2] + E[(E[\hat{f}(x_0)] - \hat{f}(x_0))^2]$

$= [\text{Bias}(\hat{f}(x_0))]^2 \qquad = \text{Var}(\hat{f}(x_0))$

Bias - Var Tradeoff



Double Descent Phenomenon



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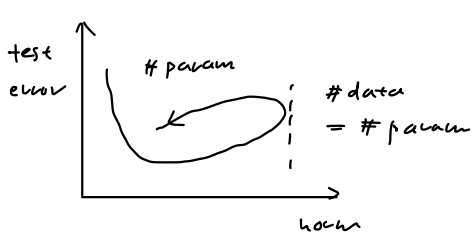
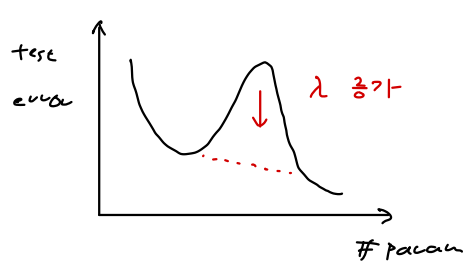
norm $\|\theta\|$ $\Rightarrow$ Double Descent Phenomenon $\Downarrow$



Linear Regression

$\Rightarrow X^T y = X^T X \theta + \lambda I$



- training error $\hat{\varepsilon}(h) = \frac{1}{n} \sum \mathbb{1}\{h(x^{(i)}) \neq y^{(i)}\}$

- generalization error $\varepsilon(h) = P_{(x,y)\sim D}(h(x) \neq y)$

Lemma (i) $P(A_1 \cup A_2 \cdots \cup A_k) \leq P(A_1) + \cdots + P(A_k)$

(ii) Chernoff bound
$$\begin{cases} Z_1 \ldots Z_n \overset{i.i.d}{\sim} \text{Bernoulli}(\phi) \\ \hat{\phi} = \frac{1}{n} \sum Z_i \end{cases}$$

$$\Rightarrow P(|\phi - \hat{\phi}| > \gamma) \leq 2e^{-2\gamma^2 n}$$

Hypothesis class $\mathcal{H} = \{h_1, \ldots, h_k\}$

then
$$P(|\varepsilon(h_i) - \hat{\varepsilon}(h_i)| > \gamma) \leq 2e^{-2\gamma^2 n}$$
$$P(\exists h \text{ s.t } |\varepsilon(h) - \hat{\varepsilon}(h)| > \gamma) \leq \underbrace{2ke^{-2\gamma^2 n}}_{=\delta} \quad \text{i.e} \quad \gamma = \sqrt{\frac{1}{2n}\log\left(\frac{2k}{\delta}\right)}$$

최소 $1-\delta$의 확률로 $|\varepsilon(h) - \hat{\varepsilon}(h)| \leq \gamma$

- $\varepsilon(\underbrace{\hat{h}}_{\text{training}}) \leq \hat{\varepsilon}(\hat{h}) + \gamma \leq \hat{\varepsilon}(h^*) + \gamma \leq \hat{\varepsilon}(\underbrace{h^*}_{\text{generalization}}) + 2\gamma$

- $\varepsilon(\hat{h}) \leq \hat{\varepsilon}(h^*) + 2\sqrt{\frac{1}{2n}\log\frac{2k}{\delta}}$

- $2\sqrt{\frac{1}{2n}\log\frac{2k}{\delta}} < 2\gamma$ then $n \geq \frac{1}{2\gamma^2}\log\left(\frac{2k}{\delta}\right) = O\left(\frac{1}{\gamma^2}\log\frac{k}{\delta}\right)$